Machine Intelligence II
Lecture Notes

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1 Mathematical Synthesis: Spectral Structure and Online Learning

Machine Intelligence II begins from unsupervised structure: what can be learned when the data arrive without labels? Principal component analysis is the clean prototype because every step is visible. Center the data, measure covariance, diagonalize the covariance, project onto the strongest directions, and track exactly what variance was kept or discarded. Hebbian learning then asks how the same directions can be found online, one sample at a time.

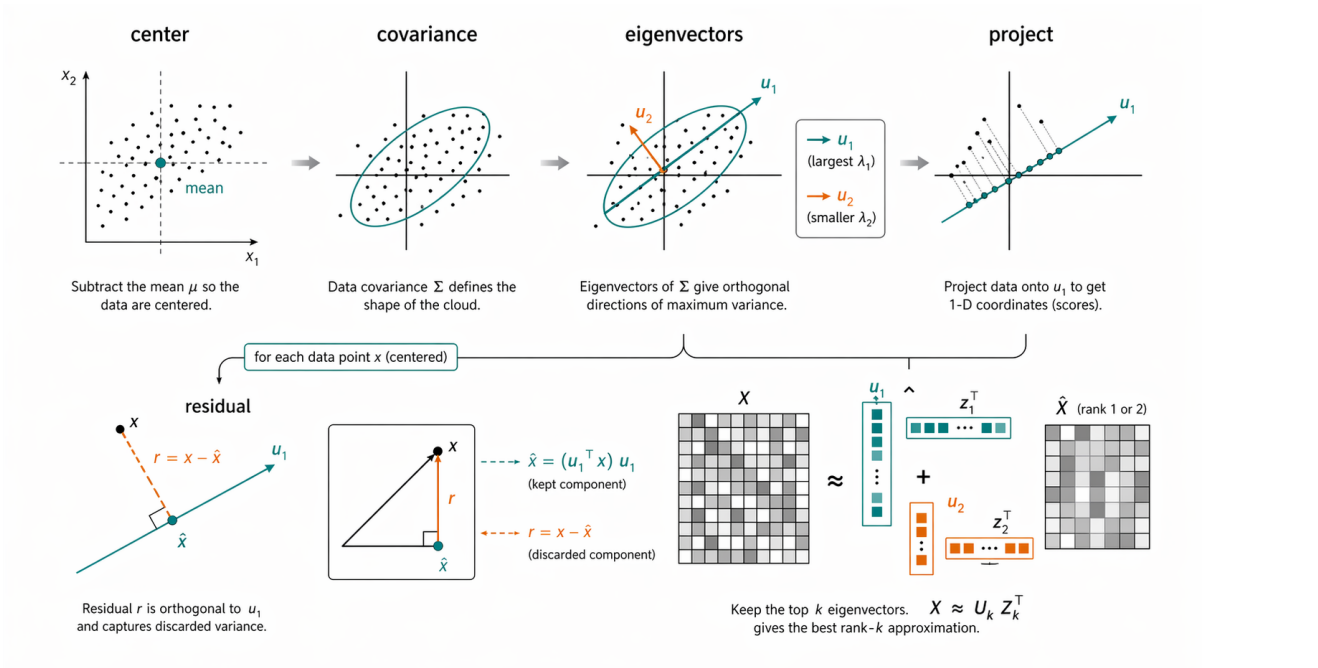

Figure 1: PCA as spectral compression. Centering removes the mean, the covariance ellipse reveals directions of variation, eigenvectors provide orthogonal coordinates, projection keeps high-variance coordinates, and residuals are the discarded orthogonal variation.

1.1 The Covariance Matrix Is A Geometry

Let centered data vectors be rows of $X \in \mathbb{R}^{n \times d}$. The empirical covariance is

$$\Sigma = \frac{1}{n} X^\top X.$$

For any unit direction u, the variance of the data after projection onto u is

$$\frac{1}{n} \|Xu\|^2 = u^\top \Sigma u.$$

So PCA asks for the direction with maximal quadratic form:

$$u_1 \in \arg\max_{\|u\|=1} u^\top \Sigma u.$$

The solution is the leading eigenvector of Σ. The corresponding eigenvalue is the variance captured along that direction.

1.2 Projection And Reconstruction

If $U_k = [u_1, \ldots, u_k]$ contains the top k orthonormal eigenvectors, then the low-dimensional score vector for a data point x is

$$z = U_k^\top x,$$

and the rank-k reconstruction is

$$\hat{x} = U_k z = U_k U_k^\top x.$$

The residual is

$$r = x - \hat{x}.$$

It is orthogonal to the retained subspace, so the reconstruction separates data into an explained component and a discarded component.

Variance captured.

$$\text{captured variance}(k) = \sum_{j=1}^{k} \lambda_j, \qquad \text{fraction captured} = \frac{\sum_{j=1}^{k} \lambda_j}{\sum_{j=1}^{d} \lambda_j}.$$

The spectrum $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d$ is a diagnostic for intrinsic dimension. A fast spectral decay means the data are close to a low-dimensional subspace.

1.3 Online Learning Turns Spectra Into Dynamics

For a linear neuron $y = w^\top x$, the basic Hebbian update is

$$\Delta w = \varepsilon y x.$$

Averaged over centered samples, this becomes

$$\mathbb{E}[\Delta w] = \varepsilon \mathbb{E}[x x^\top] w = \varepsilon C w.$$

So Hebbian learning is a stochastic power method. Components of w along high-eigenvalue directions grow faster than components along low-eigenvalue directions. The direction of w therefore approaches the first principal component, but the norm grows without bound unless the learning rule includes a constraint.

Oja's rule adds that constraint:

$$\Delta w = \varepsilon y (x - y w).$$

The first term is Hebbian attraction toward directions that explain variance. The second term is a local normalization pressure. Together they make the unit vector $w/\|w\|$ converge to the leading principal direction.

1.4 Why PCA Is The Right Baseline

PCA is linear, unsupervised, and global. Those limitations are also why it is such a useful baseline. If a nonlinear representation method improves on PCA, it should be clear which PCA assumption it breaks: linearity, globality, Euclidean reconstruction error, or Gaussian-like covariance structure.

The mathematical virtue of PCA is that it solves two problems at once. It maximizes retained variance and minimizes squared reconstruction error:

$$\min_{\operatorname{rank}(\hat{X})\le k} \|X - \hat{X}\|_F^2.$$

By the Eckart–Young theorem, the solution is the truncated singular value decomposition. This makes PCA the reference point for later factor models, autoencoders, manifold learning, and probabilistic latent-variable models.

2 Lecture 11: Principal Component Analysis

Principal Component Analysis (PCA), also known as the Karhunen–Loeve transform, is the first projection method in Machine Intelligence II. The problem starts with observations

$$x^{(\alpha)} \in \mathbb{R}^N, \qquad \alpha = 1, \dots, p,$$

and asks which directions in feature space expose the relevant structure of the data. PCA answers this question with a variance criterion: directions are interesting when the data vary strongly along them.

2.1 Projection Methods and Data Structure

The lecture contrasts two families of unsupervised methods. Projection methods search for informative directions in feature space, while clustering methods search for groups, categories, or prototypes. PCA belongs to the projection family. It does not first ask which class an observation belongs to; it asks which coordinate system makes the data cloud easiest to understand.

From raw variables to directions. An elementary feature is one coordinate axis of the original measurement space. A complex feature is a direction e_a in that same space. Its value on an observation is the scalar projection

$$u_a(x) = e_a^\top x.$$

PCA is a principled way to choose such directions from the data.

The iris example in the slides is a useful starting intuition. A flower is measured by several numerical attributes, and scatter plots of pairs of attributes already show some structure. PCA generalizes this search beyond pairwise plots: it finds linear combinations of the original attributes that can reveal separation, compression, or nuisance variation.

2.2 Moments of the Data Cloud

PCA is built from the first two empirical moments of the observations. The sample mean is the center of mass,

$$m = \frac{1}{p}\sum_{\alpha=1}^{p} x^{(\alpha)}.$$

The covariance matrix records the centered second moments:

$$C_{ij} = \frac{1}{p}\sum_{\alpha=1}^{p} \big(x_i^{(\alpha)} - m_i\big)\big(x_j^{(\alpha)} - m_j\big).$$

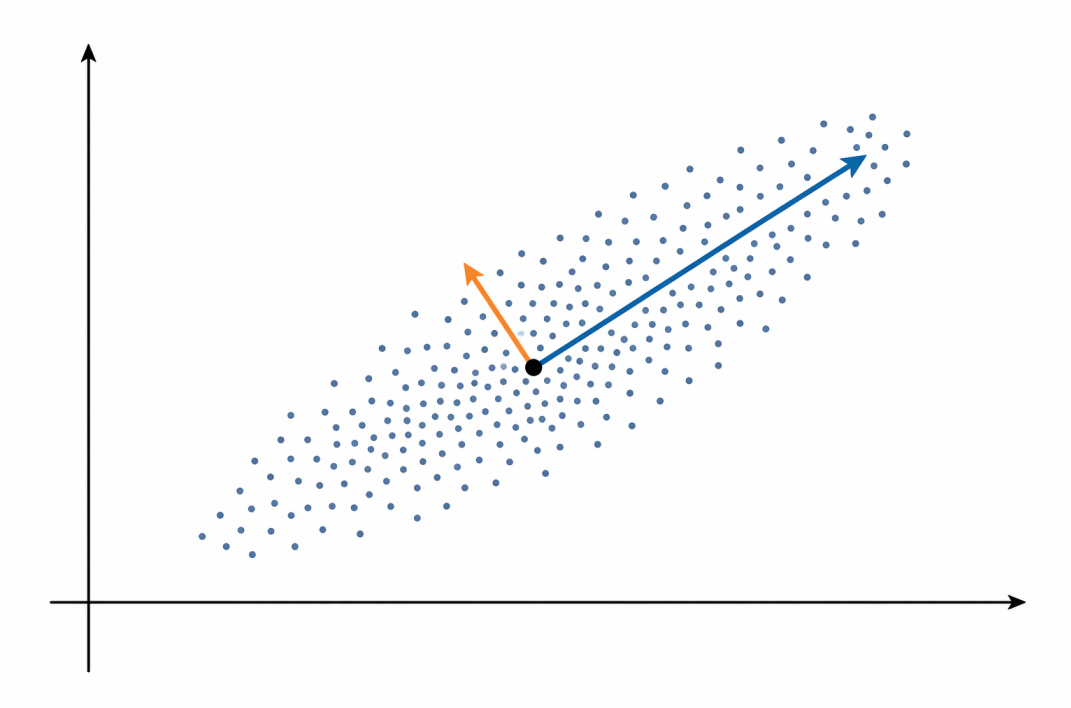

Figure 2: A data cloud has different variances in different directions. PCA chooses the first direction along the largest spread of the centered cloud, then chooses orthogonal directions that explain the remaining variance.

For centered data, where $m = 0$, this simplifies to

$$C_{ij} = \frac{1}{p} \sum_{\alpha=1}^{p} x_i^{(\alpha)} x_j^{(\alpha)}.$$

The diagonal entries C_{ii} are variances of individual variables. The off-diagonal entries C_{ij} are covariances between variables. The matrix is symmetric, so $C_{ij} = C_{ji}$. A zero covariance means two variables are uncorrelated, but it does not prove statistical independence; nonlinear dependence can remain invisible to a second-order method.

2.3 Variance Along a Complex Feature

Let e_a be a unit vector in feature space. The induced feature value is

$$u_a^{(\alpha)} = e_a^{\mathsf{T}} x^{(\alpha)}.$$

Its mean is

$$m_a = \frac{1}{p} \sum_{\alpha=1}^{p} u_a^{(\alpha)} = e_a^{\mathsf{T}} m.$$

Its variance is

$$\sigma_a^2 = \frac{1}{p} \sum_{\alpha=1}^{p} \left(u_a^{(\alpha)} - m_a\right)^2 = e_a^{\mathsf{T}} C e_a.$$

This identity is the hinge of the lecture: the covariance matrix determines the variance of the data along every possible direction. Once C is known, PCA can compare all candidate directions without returning to the raw data points.

2.4 The Principle of Maximal Variance

The first principal component is the unit direction with maximal projected variance:

$$e_1 = \arg\max_{\|e\|=1} e^\mathsf{T} C e.$$

The unit-length constraint is essential. Without it, the variance could be made arbitrarily large by scaling e. Introducing a Lagrange multiplier gives

$$\mathcal{L}(e, \lambda) = e^\mathsf{T} C e - \lambda(e^\mathsf{T} e - 1).$$

Setting the derivative with respect to e to zero yields

$$Ce = \lambda e.$$

Thus the principal components are the normalized eigenvectors of the covariance matrix. The variance along a principal component is the corresponding eigenvalue:

$$\sigma_a^2 = e_a^\mathsf{T} C e_a = \lambda_a.$$

PCA eigenproblem.

$$Ce_a = \lambda_a e_a, \qquad \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N.$$

The eigenvectors give the principal directions. The eigenvalues rank those directions by explained variance.

2.5 Properties of Principal Components

Because the covariance matrix is real and symmetric, its eigenvectors form an orthonormal basis:

$$e_i^\mathsf{T} e_j = \delta_{ij}.$$

Ordering the eigenvalues from largest to smallest orders the corresponding eigenvectors from most variable to least variable. The first principal component points along the largest variance in the full space. The second points along the largest remaining variance in the subspace orthogonal to the first, and so on.

2.5.1 Optimal Linear Dimensionality Reduction

Any observation can be represented in the principal-component basis:

$$x = a_1 e_1 + a_2 e_2 + \cdots + a_N e_N, \qquad a_j = e_j^\mathsf{T} x.$$

Keeping only the first M terms gives the rank-M PCA approximation

$$\tilde{x} = a_1 e_1 + a_2 e_2 + \cdots + a_M e_M.$$

Among all M-dimensional linear projections, this choice minimizes the mean-squared reconstruction error:

$$E = \frac{1}{p} \sum_{\alpha=1}^{p} \|x^{(\alpha)} - \tilde{x}^{(\alpha)}\|^2 = \sum_{j=M+1}^{N} \lambda_j.$$

In words, the error is the variance left behind in the discarded principal components.

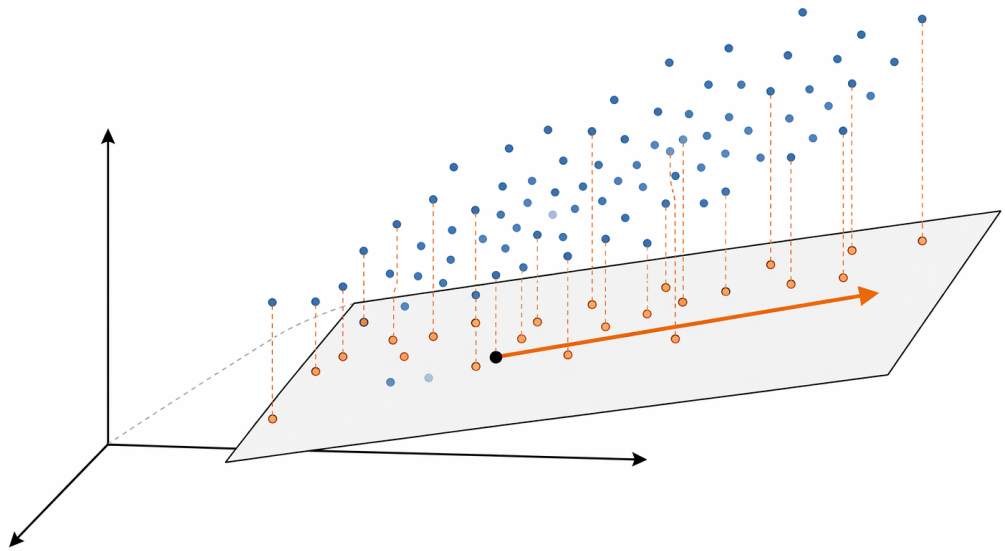

Figure 3: Dimensionality reduction by projection. PCA keeps the principal subspace that preserves the largest variance and leaves only the orthogonal residual as reconstruction error.

2.5.2 Decorrelation

Let

$$M = (e_1, e_2, \ldots, e_N)$$

be the matrix whose columns are the principal components. In this basis, the covariance matrix is diagonal:

$$M^\top C M = \Lambda = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_N).$$

The transformed features are therefore uncorrelated. This is why PCA is often used before regression, classification, clustering, or visualization: it rotates the coordinate system into axes that separate second-order variation cleanly.

2.5.3 Whitening

Variance is scale sensitive. If one input dimension is measured in much larger units than another, a variance criterion may prioritize scale rather than structure. PCA therefore only makes direct sense when feature scales are comparable, or after suitable normalization.

After decorrelation, whitening rescales every principal direction to unit variance:

$$v^{(\alpha)} = \Lambda^{-1/2} M^\top x^{(\alpha)}.$$

The result has covariance equal to the identity matrix. Whitening removes both linear correlations and unequal variances, making the transformed representation scale-standardized.

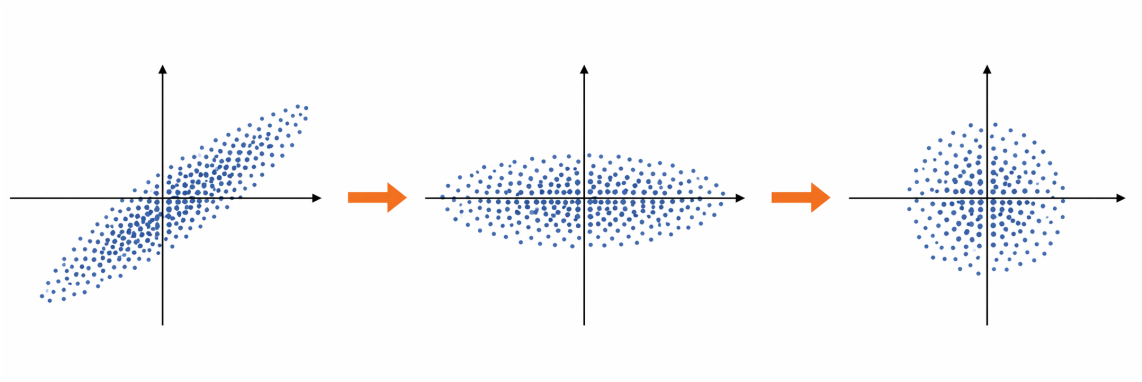

Figure 4: PCA first rotates an elongated data cloud into its eigenbasis, where the covariance is diagonal. Whitening then rescales the axes so the transformed cloud has unit variance in every retained direction.

2.6 Applications and Interpretations

2.6.1 Iris Data

The iris slides use a familiar biological dataset to motivate projection methods. Pairwise scatter plots can show partial separation between setosa, versicolor, and virginica, but PCA asks for the strongest directions in the full feature space rather than inspecting every pair of raw features. This makes it a compact exploratory tool when there are more variables than can be inspected directly.

2.6.2 Leptograpsus Crab Measurements

The Leptograpsus example is more explicitly about complex features. Each crab is described by five measurements: frontal lobe size, rear width, carapace length, carapace width, and body depth. PCA forms linear combinations of these elementary features. The leading component can capture overall size, while later components may align with sex- or species-related variation if those factors explain substantial covariance structure.

This example also warns against overinterpreting components. A principal component is a direction chosen by variance, not by semantics. A component may become interpretable as “size” or “sex” only after examining its loadings and how observations project onto it.

2.6.3 Outlier Detection

The lecture also points out a less obvious use of PCA: components with the smallest eigenvalues can help detect outliers or novel features. A normal data point should not vary much along a low-variance direction. A large projection onto such a direction can therefore indicate an unusual observation, a measurement artifact, or a rare structure not captured by the dominant modes.

2.7 Computational Notes and Extensions

PCA is linear, second-order, and variance-based. This is both its strength and its limitation. It is efficient, interpretable, and useful for preprocessing, dimensionality reduction, and compression. But very large covariance matrices can be numerically unstable or expensive, and second-order structure cannot capture every kind of dependence.

The slide summary points to two algorithmic routes for extracting leading principal components: Hebbian learning and the power method. Both are useful when the largest eigenvectors matter more than the full eigenspectrum. It also points forward to extensions. Kernel PCA replaces the raw feature space with nonlinear features, while probabilistic PCA and factor analysis add an explicit generative model. Later methods such as ICA change the objective from decorrelation to statistical independence.

Summary. PCA begins with the covariance matrix and asks which unit directions maximize projected variance. The solution is the eigenbasis of C: eigenvectors are principal components, and eigenvalues measure explained variance. Keeping the leading components gives the optimal linear low-dimensional reconstruction, rotating into the eigenbasis decorrelates the features, and whitening additionally rescales them to unit variance. The method is simple and powerful, but it remains linear and variance-based, so later MI2 methods relax or extend these assumptions.

3 Lecture 12: Online PCA and Hebbian Learning

The second lecture starts from the PCA eigenproblem and asks a more biological and algorithmic question: can the leading principal directions be learned incrementally, without first forming and diagonalizing the full covariance matrix?

The answer is yes, but each part of that statement matters.

A *linear neuron* is just a weighted sum $y = w^\mathsf{T} x$. It turns a high-dimensional input x into one number y, the projection of the input onto the current weight direction w. If w points along a high-variance direction of the data, then the output y varies a lot across examples.

A *Hebbian update* strengthens the weights in the direction of inputs that already produce a large output. Averaged over many samples, this update multiplies the weight vector by the covariance matrix. That is why Hebbian learning resembles the *power method*: the power method repeatedly multiplies a vector by a matrix, and repeated multiplication by a covariance matrix amplifies the eigenvector with the largest eigenvalue.

The plain Hebbian rule has one missing ingredient: it controls direction but not length. The weight vector keeps growing. Oja's rule adds a normalization term, so the direction can rotate toward the first principal component while the length stays bounded. Sanger's rule repeats the same idea with several neurons and subtracts off directions already learned, so later neurons learn later principal components. Reversing the sign gives anti-Hebbian learning: directions with large variance are suppressed, leaving small-variance directions that can be used as novelty filters.

3.1 PCA As A Learning Problem

For centered observations $x^{(\alpha)} \in \mathbb{R}^N$, the covariance matrix is

$$C = \frac{1}{p} \sum_{\alpha=1}^{p} x^{(\alpha)} x^{(\alpha)\mathsf{T}}.$$

Centered means that the sample mean has already been subtracted, so the origin represents the average input. Each term $x^{(\alpha)} x^{(\alpha)\mathsf{T}}$ is an outer product. It records which coordinates of the input tend to be large together. The covariance matrix is therefore a summary of the shape of the data cloud.

PCA asks for the unit vector e that maximizes projected variance:

$$e_1 = \arg\max_{\|e\|=1} e^\top C e.$$

The phrase *projected variance* means: choose a direction e, drop every data point onto that line, and measure how spread out the resulting one-dimensional numbers are. PCA chooses the line where that spread is largest.

The stationary condition is the eigenvalue equation

$$Ce = \lambda e.$$

An eigenvector is a direction that the matrix C stretches but does not rotate. The eigenvalue λ is the amount of stretch. For a covariance matrix, that stretch is variance along the corresponding direction. So the first principal component is the eigenvector with the largest eigenvalue.

The offline solution computes C and diagonalizes it. Online PCA instead sees one pattern at a time and updates a weight vector. The data stream is turned into a dynamical system: as examples arrive, the vector w moves. The goal is to design the movement so that the stable direction of w is the first principal component.

3.2 A Linear Neuron

The model neuron in the lecture is deliberately simple:

$$y = w^\top x.$$

This is not meant to be a detailed biophysical neuron. It is the minimal algebraic model needed for PCA. The input x is a centered observation, w is the synaptic weight vector, and y is the scalar output. The output is large when the input points in a similar direction to the current weight vector.

This model is useful because the same expression appears in PCA. The projection of x onto a candidate direction w is exactly $w^\top x$. Learning w is therefore learning a projection direction.

3.3 Hebb's Rule Finds The Dominant Direction

Hebbian learning is summarized by the phrase: units that are active together should strengthen their connection. For the linear neuron, the update is

$$\Delta w = \varepsilon y x,$$

or componentwise,

$$\Delta w_j = \varepsilon y x_j.$$

The learning rate $\varepsilon > 0$ is small. Each sample nudges the weights in the direction of the current input, scaled by the output. Inputs that already align with w produce larger outputs and therefore larger changes.

This rule is local in the neural sense: the change in weight w_j depends only on the presynaptic activity x_j, the postsynaptic activity y, and the learning rate. No global covariance matrix has to be stored in the neuron.

To see why this is PCA, average the update over many centered samples:

$$\mathbb{E}[\Delta w] = \varepsilon \mathbb{E}[yx] = \varepsilon \mathbb{E}[xx^\mathsf{T}]w = \varepsilon C w.$$

The mean update is multiplication by the covariance matrix. This is the key bridge to PCA.

The power method is a simple eigenvector algorithm. Start with almost any vector w, repeatedly apply the matrix C, and renormalize:

$$w \leftarrow \frac{Cw}{\|Cw\|}.$$

The component of w along the largest-eigenvalue direction is stretched the most, so after many steps the vector points toward the first eigenvector. Hebbian learning is a noisy, sample-by-sample version of the same idea: it does not apply C exactly, but its average update is Cw.

Let $e_1, \ldots, e_N$ be the eigenvectors of C, with eigenvalues

$$\lambda_1 > \lambda_2 > \cdots > \lambda_N.$$

Write the weight vector in this eigenbasis:

$$w = a_1 e_1 + a_2 e_2 + \cdots + a_N e_N.$$

Since $Ce_j = \lambda_j e_j$, the averaged update gives

$$\Delta a_j = \varepsilon \lambda_j a_j.$$

The coefficient a_j says how much of w points along eigenvector e_j. The coefficient along e_1 grows fastest because λ_1 is largest. If the initial weight vector has any nonzero component in the e_1 direction, the direction of w approaches e_1.

The problem is the norm. Pure Hebbian learning does not keep $\|w\|$ fixed. The direction becomes useful, but the weight length diverges. PCA cares about directions, not arbitrary vector length, so a useful online rule must include some form of normalization.

3.4 Oja's Rule Adds The Missing Constraint

PCA is a constrained optimization problem: the direction e must have unit norm. Without the unit-norm constraint, the objective $e^\mathsf{T} C e$ could be made arbitrarily large just by scaling e. Oja's rule builds that constraint into the learning rule:

$$\Delta w = \varepsilon y (x - yw).$$

In components,

$$\Delta w_j = \varepsilon y (x_j - y w_j).$$

The first term, $\varepsilon y x$, is the Hebbian attraction term. The second term, $-\varepsilon y^2 w$, is a stabilizing decay term that grows when the output is large. It subtracts from the current weight direction, so it prevents the vector from increasing without bound.

One way to understand Oja's rule is explicit normalization. Start with a Hebbian step,

$$\tilde{w} = w + \varepsilon y x,$$

then normalize,

$$w_{\text{new}} = \frac{\tilde{w}}{\|\tilde{w}\|}.$$

For small ε, a Taylor expansion gives the Oja update to first order:

$$w_{\text{new}} - w \approx \varepsilon y(x - yw).$$

Here "to first order" means that terms of size ε^2 and smaller are ignored. When the learning rate is small, those terms are much smaller than the leading term. So Oja's rule is a local, online approximation to Hebbian learning plus Euclidean normalization.

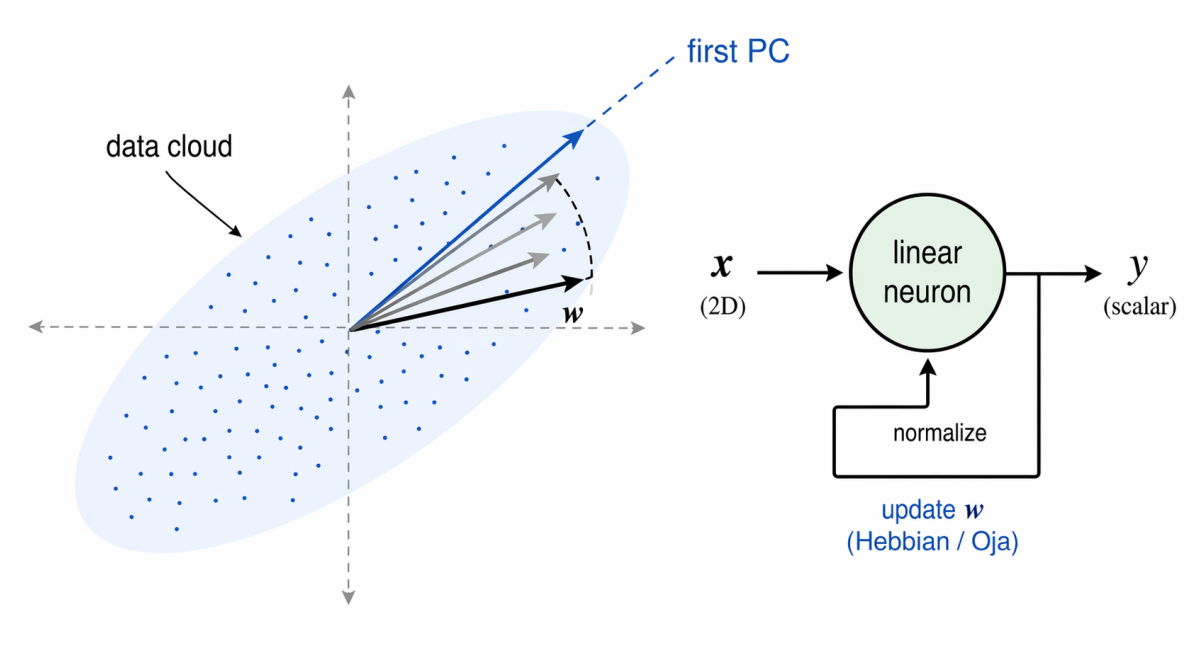

Figure 5: Online PCA with one linear neuron. Hebbian updates rotate the weight vector toward the high-variance direction of the centered data cloud. Oja's normalization keeps the vector length bounded, so the stable direction is the first principal component.

3.5 The Fixed Point Intuition

At convergence under Oja's rule, w is a unit vector aligned with the leading eigenvector. A convergence point is a direction where the average update is zero: on average, learning no longer moves the vector. This happens when Hebbian growth is exactly balanced by the normalization term. If $w = e_1$, then

$$\mathbb{E}[yx] = Cw = \lambda_1 w,$$

and

$$\mathbb{E}[y^2] = w^\top C w = \lambda_1.$$

Thus the mean update becomes

$$\varepsilon(Cw - \mathbb{E}[y^2]w) = \varepsilon(\lambda_1 w - \lambda_1 w) = 0.$$

The leading eigenvector is stable because any small component along a lower-variance direction grows more slowly than the leading component. If the vector is nudged slightly away from e_1, the dynamics amplify the e_1 part more strongly and pull the direction back.

3.6 From One Component To Many: Sanger's Rule

One neuron learns one principal direction. If several neurons all use the same Oja rule on the same input, they all have the same favorite solution: the first principal component. To learn several directions, the system needs multiple neurons and a mechanism that prevents them all from converging to the same first component.

Let

$$y_i = w_i^\mathsf{T} x$$

be the output of neuron i. Sanger's generalized Hebbian algorithm uses

$$\Delta w_i = \varepsilon y_i \left(x - \sum_{k=1}^{i} y_k w_k \right).$$

Componentwise this is

$$\Delta w_{ij} = \varepsilon y_i \left(x_j - \sum_{k=1}^{i} w_{kj} y_k \right).$$

For $i = 1$, this is Oja's rule. For $i = 2$, the input is first stripped of the component already explained by w_1. For $i = 3$, the input is stripped of the components explained by w_1 and w_2, and so on.

This is what orthogonalization means here. A later neuron does not learn from the full input; it learns from the residual, the part of the input that cannot already be reconstructed from earlier neurons. Gram-Schmidt is the classical procedure that turns a list of vectors into perpendicular directions by subtracting projections onto directions already chosen. Sanger's rule performs the same subtraction online, while data are arriving.

Each neuron performs Oja-like learning on the residual left after earlier neurons have taken their directions. Under the usual assumptions, the weights converge to the ordered principal components:

$$w_1 \to e_1, \qquad w_2 \to e_2, \qquad \ldots, \qquad w_M \to e_M.$$

After learning, the neuron outputs are the PCA scores:

$$y_i = e_i^\mathsf{T} x.$$

3.7 Novelty Filters And Anti-Hebbian Learning

Large-variance directions describe what is common. Small-variance directions can reveal what is surprising. For example, suppose normal data mostly lie near a line or a plane. Motion along that line or plane is ordinary, but motion away from it is unusual. The low-variance direction points away from the normal data shape.

The lecture expresses this with the last principal component. After PCA learning,

$$y = e_N^\mathsf{T} x$$

is the projection onto the direction with the smallest eigenvalue. A large absolute value of y signals that the input has energy in a direction where normal data vary little. This is the novelty-filter idea: the output is small for typical inputs and large when the input violates the learned regularity.

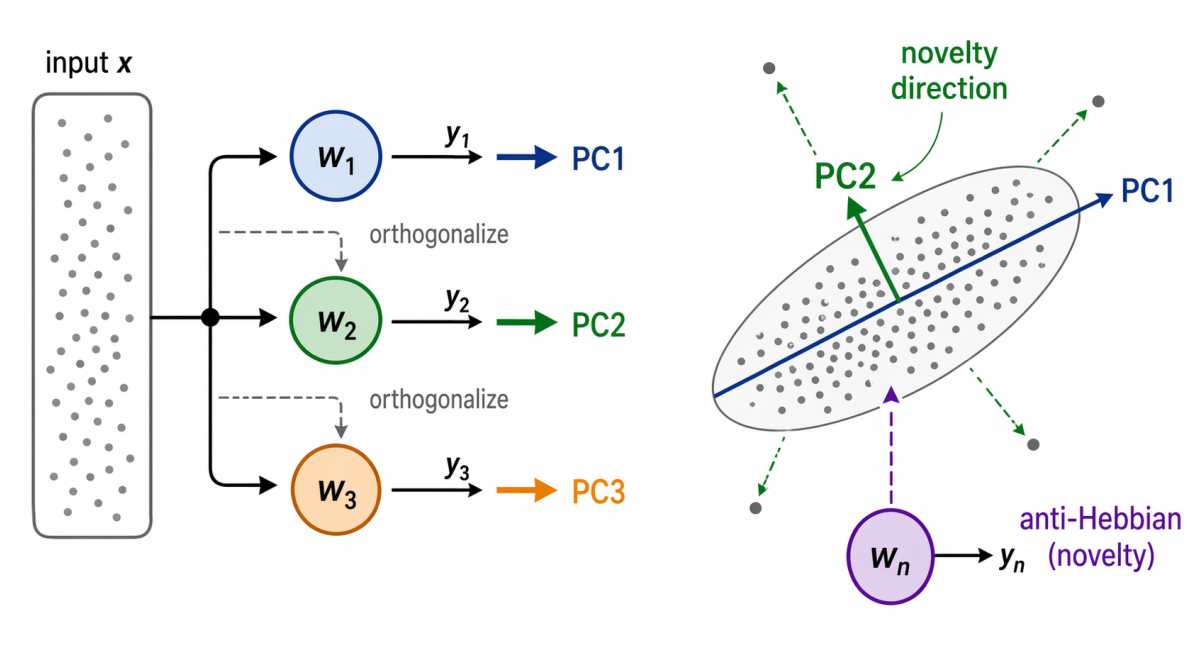

Figure 6: Multiple Hebbian neurons need competition or orthogonalization. Sanger's rule makes each later neuron learn from the residual left by earlier neurons, producing ordered principal components. Reversing the sign gives an anti-Hebbian novelty direction aligned with low-variance structure.

The online rule is anti-Hebbian:

$$\Delta w = -\varepsilon y x.$$

Anti-Hebbian means that co-activation weakens rather than strengthens a connection. If an input direction repeatedly produces a large output, the weight vector is pushed away from that direction.

Averaging gives

$$\mathbb{E}[\Delta w] = -\varepsilon C w.$$

In the eigenbasis,

$$\Delta a_j = -\varepsilon \lambda_j a_j.$$

Large-eigenvalue components decay fastest because their λ_j values are largest. With normalization, the remaining direction is the eigenvector with the smallest eigenvalue. A normalized anti-Hebbian version of Oja's rule therefore converges to e_N.

3.8 Connection To Total Least Squares

The novelty-filter direction also appears in linear regression geometry. A residual is the error left after a model tries to explain a point. Ordinary least squares minimizes vertical residuals and is appropriate when the x-coordinate is treated as exact and the noise is only in the dependent variable. If both coordinates are noisy, a more symmetric objective is total least squares: minimize orthogonal distances to a line.

For centered two-dimensional data, the normal vector w to the best-fit line solves

$$\min_{\|w\|=1} w^{\mathsf{T}} C w.$$

This is the opposite of the first PCA objective. Maximizing $w^\top Cw$ gives the high-variance direction along the data. Minimizing it gives the low-variance normal direction. The solution is the normalized eigenvector of C with the smallest eigenvalue.

3.9 Summary

Hebbian learning turns PCA from an offline spectral computation into an online dynamical process. Pure Hebbian learning performs covariance multiplication and therefore points toward the first principal component, but its norm diverges. Oja's rule adds normalization and stabilizes the first component. Sanger's rule adds sequential orthogonalization so multiple neurons learn ordered principal components. Anti-Hebbian learning reverses the direction of the dynamics and, with normalization, finds low-variance novelty directions.